



Statistics
math 414

Lecture 12

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Large sample theory for maximum likelihood estimates

Theorem. Under some assumptions (to be stated),
if θ_0 is the true parameter then

$$\lim_{n \rightarrow \infty} P_{\theta_0} [L(\theta_0, \underline{x}) > L(\theta, \underline{x})] = 1$$

for all $\theta \neq \theta_0$

(This says that asymptotically the likelihood function is maximized at the true value.)

A sketch of proof

The inequality $L(\theta_0; \underline{x}) > L(\theta; \underline{x})$
is equivalent to

$$\sum \log f(x_i; \theta_0) > \sum \log f(x_i; \theta)$$

$$\Leftrightarrow \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} < 0$$

$$\Leftrightarrow \frac{1}{n} \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} < 0.$$

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moreover, this could only happen if $f(x_i; \theta) \neq f(x_i; \theta_0)$
for $\theta \neq \theta_0$.

A sketch of proof

in
probability

$$\frac{1}{n} \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)}$$



$$E_{\theta_0} \left[\log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

A sketch of proof

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$$E_{\theta_0} \left[\log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

Assumption that this
exists.



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Assumption that this
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$$\leq \log \left[E_{\theta_0} \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

A sketch of proof

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$$\frac{1}{n} \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)}$$



$$E_{\theta_0} \left[\log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

Assumption that this
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By Jensen's inequality



$$\leq \log \left[E_{\theta_0} \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

A sketch of proof

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$$\frac{1}{n} \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)}$$



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$$\leq \log \left[E_{\theta_0} \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$


$$E_{\theta_0} \left[\frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right] = \int \frac{f(x; \theta)}{f(x; \theta_0)} \cdot f(x; \theta_0) dx = 1$$

A sketch of proof

in
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$$\frac{1}{n} \sum_{i=1}^n \log \frac{f(x_i; \theta)}{f(x_i; \theta_0)}$$



$$E_{\theta_0} \left[\log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

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$$\leq \log \left[E_{\theta_0} \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

$$= 0$$


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since $\log 1 = 0$

A sketch of proof

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$$E_{\theta_0} \left[\log \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

Assumption that this
exists.

By Jensen's inequality

$$\leq \log \left[E_{\theta_0} \frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right]$$

$= 0$

$$E_{\theta_0} \left[\frac{f(x_i; \theta)}{f(x_i; \theta_0)} \right] = \int \frac{f(x; \theta)}{f(x; \theta_0)} \cdot f(x; \theta_0) dx = 1$$

This requires an
Assumption that the p.d.f. have common
domains.

since $\log 1 = 0$

Based on this theorem, we can argue that the value θ that maximizes $l(\theta; \underline{x})$ is close to θ_0 . This can be converted into a proof of the consistency of the maximum likelihood estimate, i.e., if $\hat{\theta}_n$ maximizes $l(\theta; \underline{x})$ then $\hat{\theta}_n \xrightarrow{\text{in probability}} \theta_0$ as $n \rightarrow \infty$.

We will not cover the proof.